

1)Implementation of DECISION TREE

ALGORITHM:

CODE:

```
import pandas as pd

from sklearn.preprocessing import
LabelEncoder

from sklearn.tree import DecisionTreeClassifier


# Student dataset

data = {

    'Study_Hours': ['High', 'High', 'Medium', 'Low',
'Low',

                    'Medium', 'High', 'Medium', 'Low',
'High'],

    'Attendance': ['Good', 'Poor', 'Good', 'Good',
'Poor',

                  'Poor', 'Good', 'Good', 'Poor', 'Good'],

    'Assignment': ['Yes', 'Yes', 'No', 'Yes', 'No',
                  'Yes', 'Yes', 'No', 'No', 'Yes'],

    'Pass': ['Yes', 'Yes', 'Yes', 'No', 'No',
            'No', 'Yes', 'Yes', 'No', 'Yes']

}


df = pd.DataFrame(data)


# Encode categorical data

encoders = {}

for col in df.columns:

    le = LabelEncoder()
```

```
df[col] = le.fit_transform(df[col])

encoders[col] = le

# Features and target

X = df[['Study_Hours', 'Attendance',
'Assignment']]

y = df['Pass']

# Train ID3 model

model =
DecisionTreeClassifier(criterion='entropy')

model.fit(X, y)

# New student data

new_student = pd.DataFrame({

    'Study_Hours': ['High'],

    'Attendance': ['Good'],

    'Assignment': ['Yes']

})

# Encode new data

for col in new_student.columns:

    new_student[col] =
encoders[col].transform(new_student[col])

# Predict

prediction = model.predict(new_student)

result =
encoders['Pass'].inverse_transform(prediction)
```

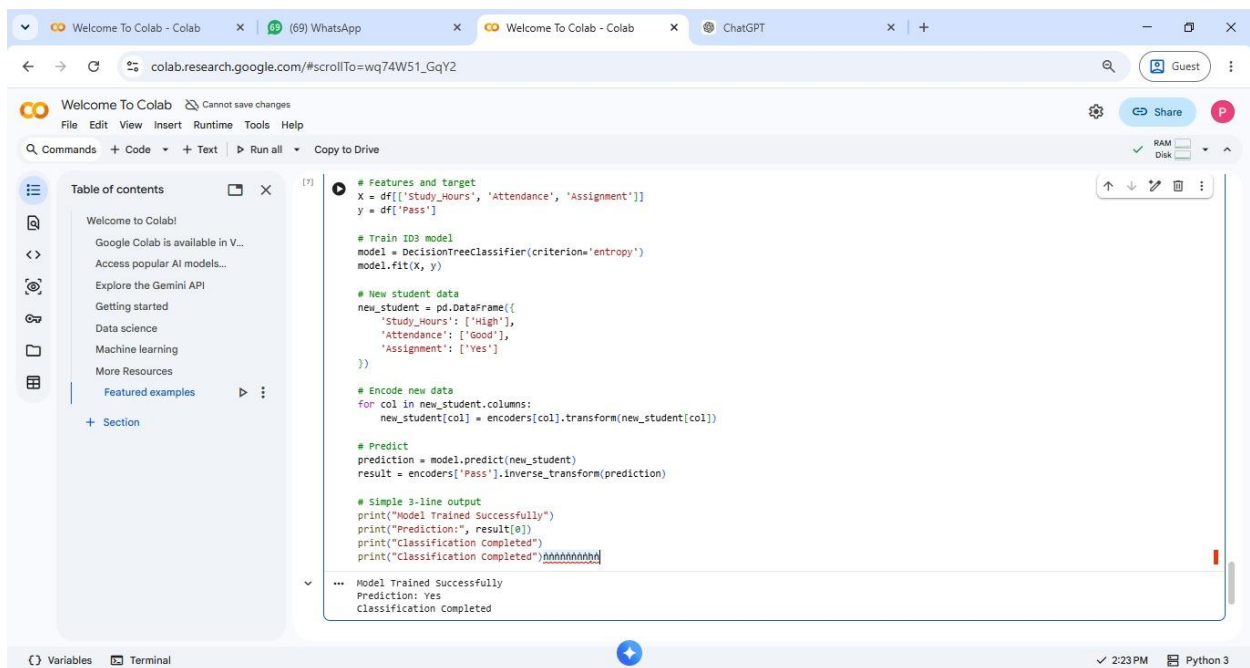
```
# Simple 3-line output
```

```
print("Model Trained Successfully")
```

```
print("Prediction:", result[0])
```

```
print("Classification Completed")
```

OUTPUT:



The screenshot shows a Google Colab notebook interface. The browser tabs at the top include 'Welcome To Colab - Colab', '(69) WhatsApp', and 'ChatGPT'. The notebook's URL is 'colab.research.google.com/#scrollTo=wq74W51_GqY2'. The notebook has a 'Welcome To Colab' header with a 'Cannot save changes' warning. The left sidebar contains a 'Table of contents' with links to 'Welcome to Colab!', 'Google Colab is available in V...', 'Access popular AI models...', 'Explore the Gemini API', 'Getting started', 'Data science', 'Machine learning', 'More Resources', and 'Featured examples'. The main code cell, labeled '[7]', contains the following Python code:

```
# Features and target
x = df[['Study_Hours', 'Attendance', 'Assignment']]
y = df['Pass']

# Train ID3 model
model = DecisionTreeClassifier(criterion='entropy')
model.fit(X, y)

# New student data
new_student = pd.DataFrame({
    'Study_Hours': ['High'],
    'Attendance': ['Good'],
    'Assignment': ['Yes']
})

# Encode new data
for col in new_student.columns:
    new_student[col] = encoders[col].transform(new_student[col])

# Predict
prediction = model.predict(new_student)
result = encoders['Pass'].inverse_transform(prediction)

# Simple 3-line output
print("Model Trained Successfully")
print("Prediction:", result[0])
print("Classification Completed")
print("Classification Completed")
```

The output of the code cell is displayed below the code:

```
Model Trained Successfully
Prediction: Yes
Classification Completed
```

The bottom status bar shows 'Variables', 'Terminal', a blue star icon, '2:23 PM', and 'Python 3'.